



2024-2025 SPRING TERM
Department of Genetics and Bioengineering
BIOE 492 - Senior Design II

Insight into the links between paternal mitochondrial elimination
(PME) mechanism and male infertility in human reproductive
genetics.

Asst. Prof. Dr. Candan Eker

By:

Joman Bashir Abumuhfouz 122201098

Nour Hariri 121201014

Farah Sbitan 121201018



Abstract

A crucial post-fertilization mechanism known as paternal mitochondrial elimination (PME) selectively breaks down male mitochondria to ensure maternal transmission of mitochondrial DNA. Male infertility, which affects over half of infertility cases globally, may be exacerbated by disruption of this mechanism. The function of PARKIN (PRKN), a crucial gene for mitophagy, in PME and its possible dysregulation in male infertility are examined in this work. DNA samples from male infertility patients and fertile controls were examined for PARKIN expression using qPCR analysis adjusted to the housekeeping gene GAPDH. Whole blood sample DNA was extracted, confirmed using Nanodrop spectrophotometry, and amplified using qPCR based on SYBR Green. A small increase of PARKIN expression was suggested by the somewhat lower average ΔC_t values found in patient samples according to the $\Delta\Delta C_t$ analysis. Fold change values in two patient samples were in line with potential copy number variation. These early results suggest a possible connection between male infertility and compromised mitochondrial clearance and altered PARKIN expression. To confirm PARKIN's diagnostic or therapeutic value in reproductive genetics, more validation is needed using larger cohorts and other markers like PINK1.



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1.Introduction

1.1 Background on Mitochondrial Inheritance

The creation of cellular energy and other metabolic functions are carried out by mitochondria, which are essential organelles. They are unique in that they have their own DNA (mtDNA), which is only inherited from the mother in the majority of eukaryotes (Cummins, 2000). During fertilization, sperm cells transfer paternal mitochondria into the oocyte; however, these are methodically removed to guarantee mitochondrial homoplasmy. This process, called paternal mitochondrial elimination (PME), avoids possible conflicts between maternal and paternal mtDNA and maintains the integrity of the mitochondrial genome. Heteroplasmy, which is associated with mitochondrial dysfunction, developmental defects, and hereditary illnesses, could arise from a failure in this system (Pacheco de Melo & Camargo, 2021).

1.2 Link Between PME and Male Infertility

A combination of genetic, physiological, and environmental variables can lead to male infertility, which accounts for 40–50% of instances worldwide (U.S. News & World Report, 2024; EAU, n.d.). Defective mitochondria are usually eliminated to preserve quality during spermatogenesis, whereas a decrease in the number of mtDNA copies takes place to improve sperm function. However, paternal mtDNA is not passed on to the embryo thanks to a different post-fertilization procedure called PME. Sperm function and the effectiveness of fertilization can be hampered by disturbances in either PME or mitochondrial remodeling (Yan et al., 2019). Determining the molecular regulation of PME brings up new possibilities for determining the genetic reasons for infertility and offers insights into the health of the male reproductive system.

1.3 Molecular Mechanisms of PME

Following fertilization, the removal of paternal mitochondria is regulated by a number of processes. Mammals undergo autophagic breakdown (mitophagy) once the ubiquitin–proteasome system designates paternal mitochondria for destruction (Sato & Sato, 2013). Important regulating genes include PRKN (Parkin), an E3 ubiquitin ligase that aids in mitochondrial clearance, and PINK1, which detects damaged mitochondria (Jin et al., 2016). It has been

suggested that other genes, including ESX1 and CCND1, affect mitochondrial dynamics during spermatogenesis and fertilization (Yu et al., 2017). These genes' malfunctions could cause PME to be delayed or compromised, which would lead to the persistence of paternal mitochondria and infertility.

1.4 The Role of PINK1 and PARKIN in Mitochondrial Elimination

The genes for PRKN (Parkin RBR E3 ubiquitin protein ligase) and PINK1 (PTEN-induced kinase 1) are essential for mitochondrial quality control, especially when it comes to mitophagy regulation. By detecting and eliminating damaged mitochondria from cells, these genes contribute to a well-preserved system that maintains cellular homeostasis and mitochondrial function.

PINK1 is quickly broken down after being transported into healthy mitochondria under normal physiological conditions. However, PINK1 builds up on the outer mitochondrial membrane and attracts cytosolic Parkin (PRKN) to the location when mitochondria lose membrane potential, a symptom of injury (Jin et al., 2016). Once located, Parkin tags the injured organelle for autophagy-lysosome destruction by ubiquitinating multiple mitochondrial surface proteins (Yu et al., 2017).

Because sperm mitochondria are vulnerable to oxidative damage, this pathway is especially crucial during spermatogenesis and fertilization. In order to identify and remove paternal mitochondria and stop their inheritance, the PINK1/Parkin pathway must be properly activated. PME failures, a higher likelihood of mitochondrial heteroplasmy, and decreased fertility have all been linked to disruption of either gene (Yan et al., 2019). Because of this, PINK1 and PRKN are both excellent candidates for research on mitochondrial inheritance problems and male infertility.

Due to technical limitations, this study focuses solely on Parkin, even though both PINK1 and Parkin are essential for PME. In particular, the qPCR parameters for Parkin and PINK1 are quite distinct, and the project's time constraints prevented both targets from being optimized. Thus, PINK1 analysis will be carried out in a later stage of the research.



1.5 GAPDH as a Reference Gene

Since Glyceraldehyde-3-phosphate dehydrogenase (GAPDH) plays a crucial role in glycolysis and is widely expressed in human tissues, it is one of the most often utilized housekeeping genes for qPCR normalization (Barber et al., 2005). GAPDH's expression is not entirely consistent, as evidenced by a large-scale study that measured its transcripts in 72 normal human tissues. The study found significant variability, with tissue types like breast and skeletal muscle differing by up to 15 times. The need for context-specific validation instead of presuming universal stability is highlighted by this diversity.

Before using GAPDH as the internal control, we empirically verified its expression throughout our particular sample set (infertile and control male DNA samples) to guarantee the study's robustness. We created a trustworthy normalization baseline for precise comparative measurement of Parkin expression by verifying its consistency.

1.6 Research Gap and Project Focus

We carried up a thorough literature review as part of our Senior Design I project, examining the biological relevance, genetic pathways, and clinical consequences of PME in relation to male infertility. There is still a dearth of experimental evidence that directly compares the gene expression levels of PME-associated genes in fertile and infertile individuals, despite the identification of numerous pathways.

We used SYBR Green I dye and primers made especially for Parkin and GAPDH to do a qPCR analysis in order to look into this. The Department of Molecular Biology and Genetics at Istanbul University supplied pre-isolated DNA samples from male infertility patients, and control DNA was extracted in-house. 8 samples for Parkin and 8 for GAPDH were used in total, with an equal number of patient and control samples. Before amplification, Nanodrop spectrophotometry was used to confirm the concentration and purity of the sample. Only Parkin expression was measured in this phase due to time constraints and gene-specific machine settings.



2. Materials and Methodology

2.1 DNA Isolation

A crucial stage in molecular biology, DNA isolation makes it possible to isolate genetic material from cells for use in PCR, qPCR, sequencing, and cloning, among other processes. Cells are broken open, proteins and other impurities are eliminated, and the DNA is purified so that it can be accurately quantified and amplified. For gene expression studies and other genetic tests to be reliable and reproducible, effective and high-quality DNA extraction is essential (Tan & Yiap, 2009).

The Cis-DNA Blood Genomic DNA Kit was used to isolate genomic DNA in accordance with the manufacturer's instructions. The 25 control group's DNA was taken from fresh whole blood samples treated with EDTA, whereas the DNA from 25 patient samples had previously been pre-isolated and stored in the freezer before the experiment.

Each control blood sample was processed in amounts of around 200 μL . Proteinase K was used to break down proteins as part of the DNA extraction process, and NBB buffer was used to lyse cells. After the lysate was applied to a Cis-DNA spin column, the DNA attached to the membrane in a specific manner. To eliminate impurities, wash procedures were carried out utilizing WBI Wash Buffer I and WBII Wash Buffer II. To optimize yield, EB Elution Buffer was heated to 50°C before DNA was eluted.

All extracted DNA was kept at 4°C for short-term usage. The average yield from 200 μL of whole blood was between 2-4 μg , with final eluate volumes ranging from 80-100 μL each sample.



Figure 2.1: samples of patients DNA

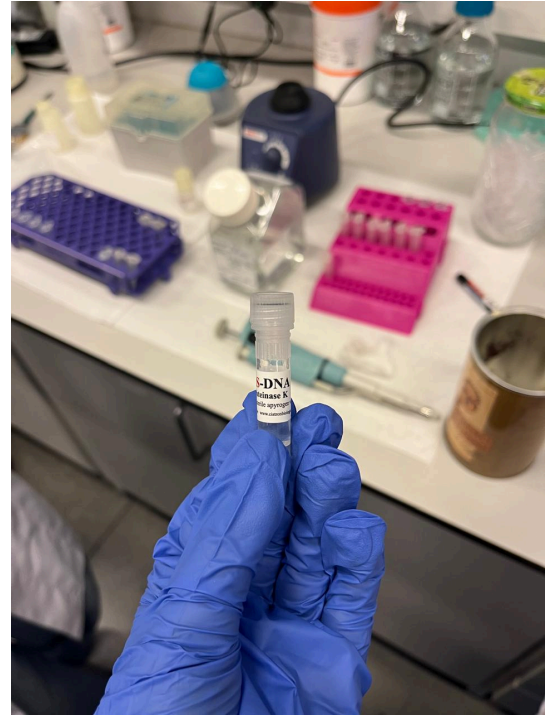


figure 2.2: Proteinase K from the Cis-DNA cell lysis during DNA isolation.

2.2 Nanodrop

A quick and accurate technique for determining the quantity and purity of nucleic acids based on UV absorbance is nanodrop spectrophotometry. It calculates DNA content and purity by measuring absorbance at 260 nm (nucleic acids) and 280 nm (proteins), using a small sample amount (usually 1-2 μ L). High-purity DNA that is appropriate for qPCR and sequencing is indicated by an A260/A280 ratio of around 1.8 (Nanodrop Thermofisher)

A Thermo Scientific NanoDrop 2000 spectrophotometer was used to measure the concentration and purity of DNA. To guarantee thorough thawing and sample homogeneity, patient DNA samples that had been previously held at -20°C were gradually brought to room temperature before being measured. To force any condensation or precipitation to the tube's bottom, the patient and control tubes were gently tapped.

The arm was lowered to start the reading after 1 μL of DNA was pipetted onto the NanoDrop pedestal for each measurement and was wiped and dried between each measurement to avoid cross-contamination. The device automatically determined the A260/A280 ratio, computed the DNA concentration (in $\text{ng}/\mu\text{L}$), and recorded absorbance values at 260 and 280 nm. For qPCR analysis later on, only samples with appropriate purity ratios were chosen.

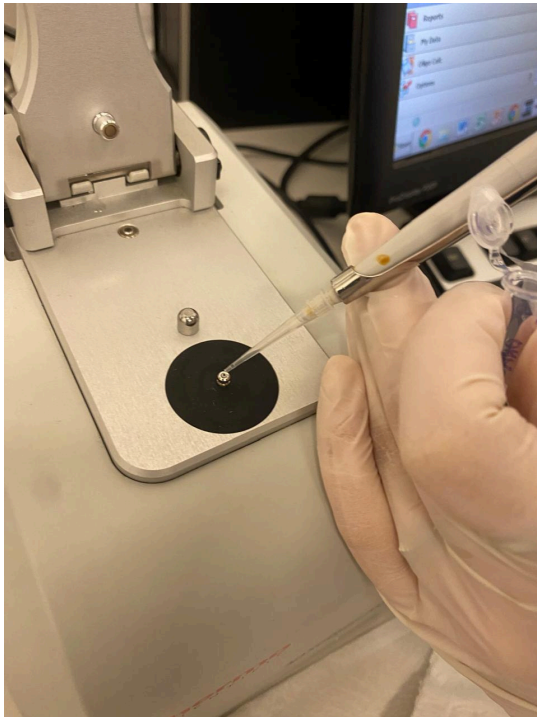


Figure 2.3: nanodrop machine

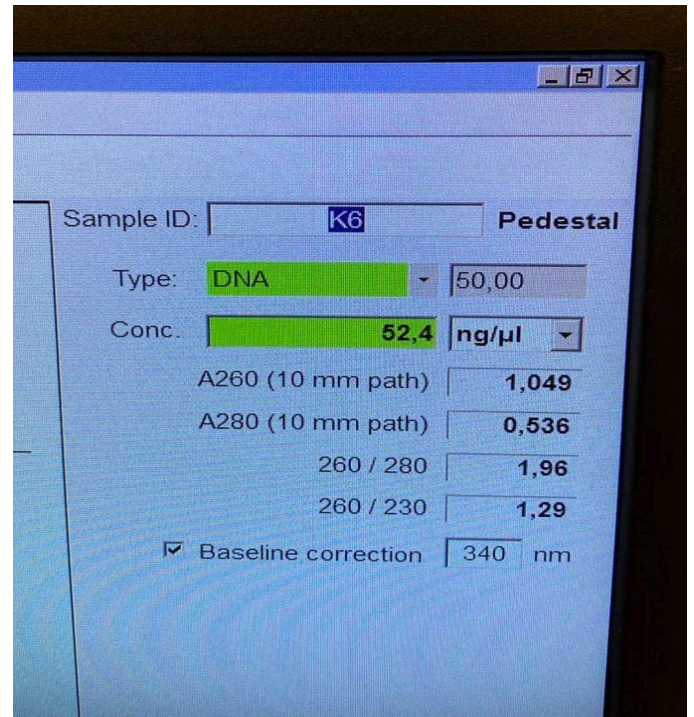


figure 2.4: nanodrop machine readings

2.3 Primer Design

We were given prepared primers in desalted solid form with a stock concentration of 100 μM for our experiment. Primers were created for two genes: the target gene of interest, PARKIN (PARK2), and the housekeeping control gene, GAPDH. We performed extensive bioinformatic validation for GAPDH primers to verify their transcript specificity, compliance with qPCR settings, and amplicon appropriateness, even though the design and synthesis were done by the supervisor Candan Eker.

Baz Dizisi 5'-3'	DMT ON/OFF	SentezNo	Ad	Sentez Skalası (nmol)	Saflık Türü	Tm (°C)	GC %	bp	MW (g/mol)	Ex. Coef	Conc. (ng/μl)	A260	Total ng	OD	Total nmol	100 μM stok - μl	50 μM stok - μl
GCACCGTCAAGGCTGAGAAC	DMTOFF	250610-1-2	GAPDH F	50	DSLT	55.9	60.00	20	6136.0	195500	178	5.7	49855	1.5	8.1	81.3	162.5
TGTTGAAGACGCCAGTGGA	DMTOFF	250610-1-3	GAPDH R	50	DSLT	53.2	57.89	19	5917.0	192600	175	5.7	49083	1.5	8.3	83.0	165.9
GTGTTTGAAGACGCCAGTGA	DMTOFF	250610-1-4	PHB2 F	50	DSLT	53.8	55.00	20	6030.0	189200	280	8.8	78396	2.4	13.0	130.0	260.0
AGAATATCAGCACTGTCCCC	DMTOFF	250610-1-5	PHB2 R	50	DSLT	53.8	55.00	20	6030.0	189200	280	8.8	78396	2.4	13.0	130.0	260.0
TCACAGTAGAGCTGTGTGC	DMTOFF	250610-1-6	AIFM1 F	50	DSLT	53.8	55.00	20	6182.0	196600	155	4.9	43336	1.3	7.0	70.1	140.2
TCATTGCAAGATTGGCTCCA	DMTOFF	250610-1-7	AIFM1 R	50	DSLT	52.4	47.62	21	6341.0	194800	195	6.0	54632	1.7	8.6	86.2	172.3
ACCACAGGCCCTCAGTACCTC	DMTOFF	250610-1-8	MYCN F	50	DSLT	58.6	59.09	22	6609.0	204200	134	4.1	37418	1.1	5.7	56.6	113.2
TGACAGAGCTGTGTGTGAGGA	DMTOFF	250610-1-9	MYCN R	50	DSLT	56.7	54.55	22	6846.0	214400	214	6.7	60044	1.8	8.5	85.3	170.6
CTTTGAGACGAGGTGTATCACC	DMTOFF	250610-1-10	STAT3 F	50	DSLT	57.1	52.17	23	7039.0	219400	116	3.6	32571	1.0	4.8	47.6	95.2
GGTCAGCATGTGTGTACACAGG	DMTOFF	250610-1-11	STAT3 R	50	DSLT	56.7	54.55	22	6775.0	214800	101	3.2	28411	0.9	4.2	41.9	83.9
TCTGAATCGGCCAAGAGGTAT	DMTOFF	250610-1-12	FIX (F9) Set 1 F	50	DSLT	53.0	45.45	22	6783.0	220300	449	14.6	125679	3.8	18.5	185.3	370.6
CATACCTGCTCCAAATTCAGTCTA	DMTOFF	250610-1-13	FIX (F9) Set 1 R	50	DSLT	52.8	36.00	25	7560.0	238600	409	12.9	114410	3.5	15.1	151.3	302.7
TCACGAGGCCAAGATTGGCA	DMTOFF	250610-1-14	FIX (F9) Set 1 F	50	DSLT	51.8	50.00	20	6175.0	202800	130	4.3	36498	1.1	5.9	59.1	118.2
TCTCCACTTCGGGGTATGCT	DMTOFF	250610-1-15	FIX (F9) Set 1 R	50	DSLT	53.8	55.00	20	6074.0	177400	430	12.6	120450	3.7	19.8	198.3	396.6
CACTCGGTGTGGACTACAG	DMTOFF	250610-1-16	INS F	50	DSLT	55.9	60.00	20	6158.0	191400	96	3.0	26963	0.8	4.4	43.8	87.6
GAGGCCCGGAGATACATCAG	DMTOFF	250610-1-17	INS R	50	DSLT	55.9	60.00	20	6096.0	196300	108	3.5	30120	0.9	4.9	49.4	98.8
GAATCTGTCGAGATGATGTGC	DMTOFF	250610-1-18	TCF7L2 F	50	DSLT	56.7	54.55	22	6775.0	216600	106	3.4	29699	0.9	4.4	43.8	87.7
TGCACTCAGCTACGACCTTGC	DMTOFF	250610-1-19	TCF7L2 R	50	DSLT	56.7	54.55	22	6646.0	195500	86	2.5	24120	0.7	3.6	36.3	72.6
TCCAGACCTCGGTGATGTA	DMTOFF	250610-1-20	HLA-DRB3 F	50	DSLT	53.8	55.00	20	6093.0	187700	251	7.7	70414	2.1	11.6	115.6	231.1
CAGATTGACAGCGTGACCT	DMTOFF	250610-1-21	HLA-DRB3 R	50	DSLT	53.8	55.00	20	6061.0	187700	539	16.7	150811	4.6	24.9	248.8	497.6
TTCCCTGGGAGAACCCGA	DMTOFF	250610-1-22	PPARG F	50	DSLT	53.8	55.00	20	6102.0	190600	132	4.1	36834	1.1	6.0	60.4	120.7
GCTTCTAGTGTCCCGCGAAT	DMTOFF	250610-1-23	PPARG R	50	DSLT	53.8	55.00	20	6084.0	181500	495	14.8	138713	4.2	22.8	228.0	456.0
CTGCAGATGACAGGTGGA	DMTOFF	250610-1-24	Parkin(PARK2) F	50	DSLT	53.8	55.00	20	6222.0	200800	79	2.5	22003	0.7	3.5	35.4	70.7
GCTTCTGCTTCGCTGATG	DMTOFF	250610-1-25	Parkin(PARK2) R	50	DSLT	53.8	55.00	20	6115.0	179500	121	3.5	33824	1.0	5.5	55.3	110.6

Figure 2.5: primer designs

2.3.1 Bioinformatics Analysis

The reference mRNA sequence for GAPDH (NM_002046.7) and the matching Ensembl transcript (ENST00000229239.10) were first obtained from NCBI. The primers were able to be compared to these basic sequences. We verified that the forward primer (GCACCGTCAAGGCTGAGAAC) and the reverse primer (TGGTGAAGACGCCAGTGTGA) both fully and precisely matched within the mature, spliced mRNA using both full primer sequences and reverse complement matching. This demonstrated that the primers did not bind to intronic or untranslated sequences since they were designed inside exon regions.

[BLAST/BLAT](#) | [VEP](#) | [Tools](#) | [BioMart](#) | [Downloads](#) | [Help & Docs](#) | [Blog](#)

Human (GRCh38.p14)

Location: 12:6,534,512-6,538,374
Gene: GAPDH

Gene-based displays

- Summary
 - Splice variants
 - Transcript comparison
 - Gene alleles
- Sequence
 - Secondary Structure
- Comparative Genomics
 - Genomic alignments
 - Gene tree
 - Gene gain/loss tree
 - Orthologues
 - Paralogues
- Ontologies
 - GO: Anatomical entity
- Phenotypes
- Genetic Variation
 - Variant table

Gene: GAPDH ENSG0000011640

Description: glyceraldehyde-3-phosphate dehydrogenase [Source:HGNC Symbol;Acc:HGNC:4141]

Gene Synonyms: GAPD

Location: [Chromosome 12: 6,534,512-6,538,374](#) forward strand.
GRCh38:CM000674.2

About this gene: This gene has 11 transcripts (splice variants), 435 orthologues and 1 paralogue.

Transcripts: [Hide transcript table](#)

Show/hide columns (1 hidden)

Transcript ID	Name	bp	Protein	Biotype	CCDS	UniProt Match	RefSeq Match
ENST00000229239.10	GAPDH-201	1285	335aa	Protein coding	CCDS8549	P04406-1	NM_002046.7

Figure 2.6: ensembl for GAPDH housekeeping gene

The Sequence Manipulation Suite: Reverse Complement Results for 19 residue sequence starting "TGGTGAAGAC".

```
TCCACTGGCGTCTTCACCA
```

Figure 2.7: forward GAPDH primer in mature mRNA

```
CGTATGACTGGGGGTGTTGGGCAGCCCTGGAGCCTTCAGTTGCAGCCATGCCTTAAGCCAGGCCAGCCTG
GCAGGGAAGCTCAAGGGAGATAAAATTCAACCTCTTGGGCCCTCCTGGGGTAAGGAGATGCTGCATTG
CCCTCTTAATGGGAGGTGGCTAGGGTGCTCACATATTCTGGAGGAGCCTCCCCTCCTATGCCTTCT
TGCCTCTTGTCTCTTAGATTGGTCGTATTGGCGCCTGGTCACCAGGCTGCTTTTAACTCTGGTAAAG
TGGATATTGTTGCCATCAATGACCCCTTCATTGACCTCAACTACATGGTGAGTGCTACATGGTGAGCCCC
AAAGCTGGTGTGGGAGGAGCCACCTGGCTGATGGGCAGCCCTTCATACCCTCACGTATTCCCCAGGTT
TACATGTTCCAATATGATTCACCCATGGCAAATTCATGGCACCCTCAAGGCTGAGAACGGGAAGCTTG
TCATCAATGGAAATCCCATCACCATCTTCCAGGAGTGAGTGGAAGACAGAATGGAAGAAATGTGCTTTGG
GGAGGCAACTAGGATGGTGTGGCTCCCTGGGTATATGGTAACCTTGTGTCCTCAATATGGTCCTGTCC
CCATCTCCCCCCCCACCCCATAGGCGAGATCCCTCCAAATCAAGTGGGGCGATGCTGGCGCTGAGTACG
TCGTGGAGTCCACTGGCGTCTTACCACCATGGAGAAGGCTGGGGTGAGTGCAAGGAGGGCCCGGGGAGG
GGAAGCTGACTCAGCCCTGCAAAGGCAGGACCCGGGTTCAACTGTCTGCTTCTGCTGTAGGCTCAT
```

Figure 2.8: reverse complement of reverse GAPDH primer

```
>NM_002046.7 Homo sapiens glyceraldehyde-3-phosphate dehydrogenase (GAPDH), transcript
variant 1, mRNA
GCTCTCTGCTCCTCCTGTTTCGACAGTCAGCCGATCTTCTTTGCGTCGCCAGCCGAGCCACATCGCTCA
GACACCATGGGGAAGGTGAAGGTCGAGTCAACGGATTGGTCGTATTGGCGCCTGGTCACCAGGGCTG
CTTTAACTCTGGTAAAGTGGATATTGTTGCCATCAATGACCCCTTCATTGACCTCAACTACATGGTTTA
CATGTTCCAATATGATTCACCCATGGCAAATTCATGGCACCGTCAAGGCTGAGAACGGGAAGCTTGTG
ATCAATGGAAATCCCATCACCATCTTCCAGGAGCGAGATCCCTCCAAATCAAGTGGGGCGATGCTGGCG
CTGAGTACGTCGTGGAGTCCACTGGCGTCTTCACCAACCATGGAGAAGGCTGGGGCTCATTGCAAGGGGG
AGCCAAAAGGGTCATCATCTCTGCCCTCTGCTGATGCCCATGTTCTGTCATGGGTGTGAACCATGAG
AAGTATGACAACAGCCTCAAGATCATCAGCAATGCCTCTGCACCAACCACTGCTTAGCACCCCTGGCCA
AGGTCATCCATGACAACCTTGGTATCTGGAAGGACTCATGACCACAGTCCATGCCATCACTGCCACCCA
GAAGACTGTGGATGGCCCTCCGGGAAACTGTGGCGTGATGGCCGCGGGGCTCTCCAGAACATCATCCCT
GCCTCTACTGGCGCTGCCAAGGCTGTGGGCAAGGTATCCCTGAGCTGAACGGGAAGCTCACTGGCATGG
CCTTCGGTGTCCCCTGCAACGTGTGAGTGGTGGACCTGACCTGCCGTCTAGAAAAACCTGCCAAATA
TGATGACATCAAGAAAGTGGTGAAGCAGGCGTGGAGGGCCCCCTCAAGGGCATCTGGGCTACACTGAG
CACCAGGTGGTCTCCTCTGACTTCAACAGCGACACCCACTCTCCACCTTTGACGCTGGGGCTGGCATTG
CCCTCAACGACCACTTGTCAAGCTCATTTCTGGTATGACAACGAATTTGGCTACAGCAACAGGGTGGT
GGACCTCATGGCCCATGGCTCCCAAGGAGTAAGACCCCTGGACCAAGCCCGCAAGAGCACAAGA
GGAAGAGAGAGACCTCACTGCTGGGAGTCCCTGCCACACTCAGTCCCCACCACTGAATCTCCCT
CCTCAGAGTTGCCATGTAGACCCCTTGAAGAGGGGAGGGGCTAGGGAGCCGACCTTGTATGTACCAT
CAATAAAGTACCCTGTGCTCAACCA
```

Figure 2.9: reverse complement of reverse GAPDH primer of mature mRNA

Using UCSC's In-Silico PCR tool, we chose the GRCh38/hg38 genome assembly and the GENCODE gene set to support this further. The program produced a single, distinct amplicon of 139 base pairs, which is within the optimal size range for SYBR Green-based qPCR (usually 70–200 bp), after simulating the qPCR process in silico. This further demonstrated that the primers avoid contaminating genomic DNA by targeting exon-exon junctions. Furthermore, melting temperature (T_m) estimations using the same UCSC tool were 63.3 °C for the forward primer and 64.0 °C for the reverse, both of which are within the ideal T_m range of 60–64 °C. In order to guarantee primer specificity and effective binding, we chose a qPCR annealing temperature of 61°C, which is around 2–3°C below the lowest T_m , based on these data.

For PARKIN, the provided forward primer was (CTGCAGATGACAGGTGGTG) and the reverse was (GCATTGTCTTCCGCTGGTG). Although these primers were used in qPCR successfully and resulted in amplification under the same thermal cycling conditions as GAPDH, their transcript-level validation could not be completed. Reverse complement analysis, direct NCBI search, and Ensembl alignment did not reveal clear matching sites within known PARKIN mRNA transcripts. This suggests either the primers span unannotated exonic regions, or they may have been designed from genomic DNA rather than mature mRNA. Regardless, additional sequence data and design context would be necessary to fully confirm their location and function. Still, the primers produced usable qPCR results, indicating that they likely bind a consistent sequence across the tested samples. Therefore, while they were included in the experimental setup, further information is needed to formally validate the PARKIN primer pair.

```
>ENST00000229239.10 GAPDH:249+387 139bp GCACCGTCAAGGCTGAGAAC GTGGTGAAGACGCCAGTGGA  
GCACCGTCAAGGCTGAGAACggaagcttgatcatcaatggaatcccatc  
accatcttcaggagcgagatccctccaaatcaagtggggcgatgctgg  
cgctgagtacgtcgtggagTCCACTGGCGTCTTACCAC
```

Figure 2.10: GAPDH qPCR Amplicon Sequence from UCSC

2.3.2 Primer Dilutions

For both genes, meticulous primer dilutions were made in terms of preparation. According to the primer datasheets and lab notes, the exact amount of nuclease-free water was used to reconstitute

the 100 μM lyophilized stock for each. For GAPDH, the forward and reverse primers received 83.6 μL and 82.4 μL of water, respectively. Then, using a 1:10 dilution of the reconstituted solutions, 10 μL of the 100 μM stock was combined with 90 μL of nuclease-free water to create 10 μM working stocks. From these, 50 μL of the forward and 50 μL of the reverse 10 μM stock solutions were combined with 150 μL of water to create 2.5 μM reaction mixes. This produced 200 μL of qPCR primer mix that was ready to use. For the PARKIN primers, the identical dilution procedure was used: 55.3 μL of water was added to the reverse primer and 35.4 μL to the 100 μM forward primer. For every qPCR experiment, they were also diluted to 10 μM and combined to produce a final primer mix of 2.5 μM . This made sure that 250 nM of each primer was introduced at the end of a 20 μL qPCR experiment.

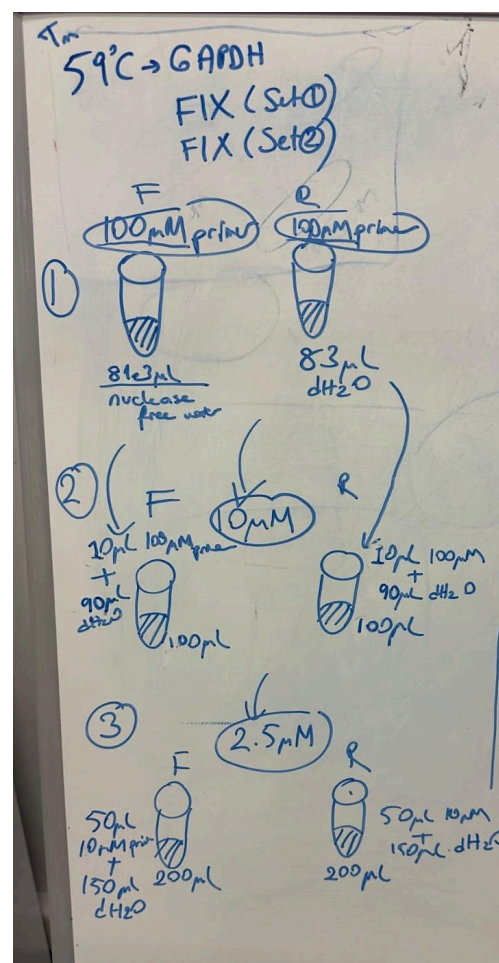
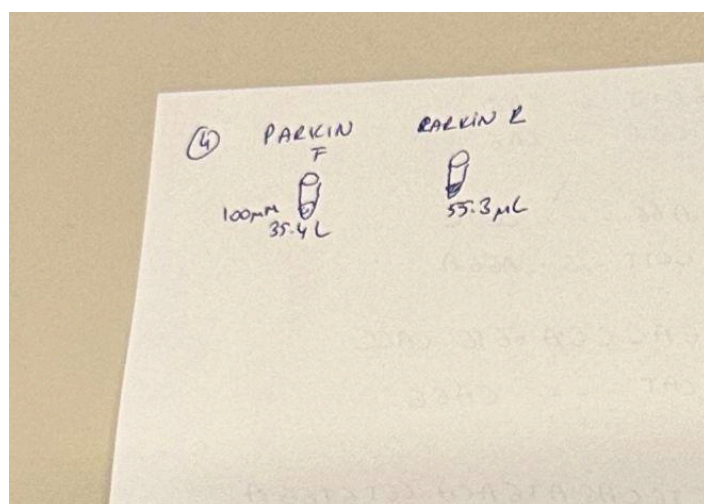


Figure 2.11: primer dilutions



2.4 Quantitative PCR (qPCR)

Real-time PCR, also referred to as quantitative PCR (qPCR), is a method that uses fluorescence to simultaneously amplify and quantify a particular DNA sequence. By monitoring fluorescence levels that rise in proportion to the amount of DNA accumulated during each cycle, qPCR offers instant feedback in contrast to conventional PCR, which necessitates post-amplification analysis via gel electrophoresis. This method is useful for gene expression analysis, diagnostic applications, and molecular biology research because it has high sensitivity, precision, and the capacity to quantify nucleic acids across a broad concentration range (Kubista et al., 2006).

To determine the relative expression levels of the target genes, qPCR with SYBR Green dye was used in this research. Accurate tracking of DNA amplification throughout PCR cycles is made possible by SYBR Green's unique attachment to double-stranded DNA and subsequent fluorescence. Particularly when analyzed using the comparative Ct ($\Delta\Delta C_t$) methodology, this method offers both qualitative gene presence detection and quantitative gene expression evaluation.

2.4.1 Reaction set up

All reagents were thawed on ice and gently vortexed before use to prepare the qPCR reactions. Depending on the instrument format, reactions were prepared either in 0.2 mL qPCR tubes or in a 96-well plate. The 20 μ L reaction mixture contained the following components:

- 10 μ L 2 $\times$ RT-PCR Master Mix (dNTPS, Taq polymerase, $MgCl_2$, and SYBR Green dye),
- 2 μ L forward primer (250 nM),
- 2 μ L reverse primer (250 nM),
- 1 μ L template DNA (containing ~50 ng total DNA),
- 5 μ L nuclease-free distilled water to reach the final volume.

The primers utilized were pre-designed and supplied to us for amplifying the target and reference genes. The primers were successful in qPCR amplification, as evidenced by the observed amplification plots and melting curves. Due to its consistent expression across samples, GAPDH was chosen as the endogenous housekeeping gene for normalization.

To avoid degradation or contamination, all components were put together on ice and in sterile conditions. To minimize pipetting variability, the reaction mix was prepared as a master mix (without template DNA) for multiple wells. After the master mix was dispensed, 1 μ L of template DNA was individually added to each reaction tube or well.

Upon preparation, the reaction mixtures were sealed and subjected to a brief spin in a mini-centrifuge at 1500 rpm for 1 minute. This was done to remove bubbles and guarantee uniform settling at the bottom of the wells. To guarantee uniform temperature distribution throughout amplification and to avert signal interference during fluorescence reading, this step is essential.

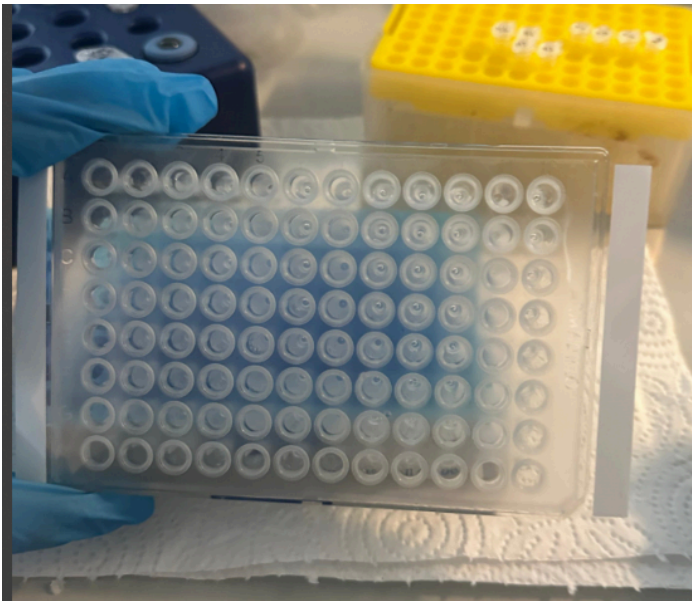


Figure 2.12: qPCR 96 well plate

Handwritten experimental layout for a 96-well qPCR plate. The layout includes gene names, sample IDs, and well numbers.

Row	Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9	Col 10	Col 11	Col 12
1 A	NTC	GAD65	GAD65	GAD65	GAD65	GAD65	GAD65	GAD65	K5	K5	K5	K5
2 B	GAD65	GAD65	GAD65	GAD65	GAD65	GAD65	GAD65	GAD65	K9	K9	K9	K9
3 C	F9	INS	GAD65	GAD65	GAD65	GAD65	GAD65	GAD65	K12	K12	K12	K12
4 D	F9	INS	GAD65	GAD65	GAD65	GAD65	GAD65	GAD65	K21	K21	K21	K21
5 E	GAD65	INS	GAD65	GAD65	GAD65	GAD65	GAD65	GAD65	K6	K6	K6	K6
6 F	GAD65	INS	GAD65	GAD65	GAD65	GAD65	GAD65	GAD65	K14	K14	K14	K14
7 G	GAD65	INS	GAD65	GAD65	GAD65	GAD65	GAD65	GAD65	K15	K15	K15	K15
8 H	GAD65	INS	GAD65	GAD65	GAD65	GAD65	GAD65	GAD65	K24	K24	K24	K24

Handwritten notes and annotations:

- Top: *2.14*, *3.7.2*, *2.15*
- Left: *Locals*, *Start*
- Right: *MYCN*, *STAT3*, *MYCN*, *STAT3*
- Bottom: *IPAT*, *CANCEL*, *done*, *cancel*, *done*

Figure 2.13: sample labeling

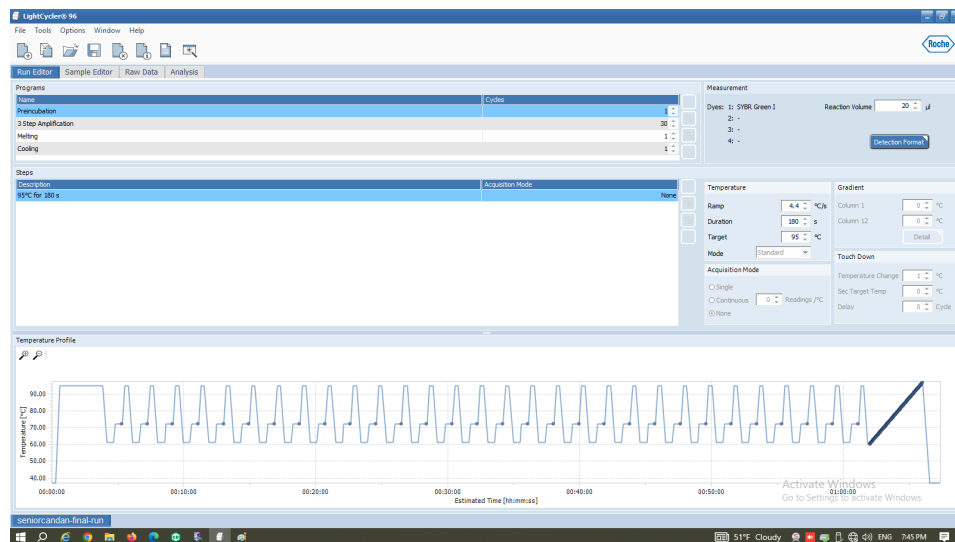


Figure 2.14: qPCR setup view

2.4.2 qPCR Thermal Cycling Protocol

We used the Roche LightCycler® instrument for the qPCR amplification. The run program was established according to standard SYBR Green protocols and was fine-tuned to identify both amplification and melting behavior of the PCR products. The complete duration of the run was 1 hour and 6 minutes.

The thermal cycling conditions were as follows:

Program	Cycle	Temperature (°C)	Duration	Acquisition Mode
Pre-incubation	1	95	180 seconds	None
Amplification	30	95	15 second	None
		61	30 seconds	None
		72	30 seconds	Single
Melting Curve	1	95	10 seconds	None



		60	10 seconds	None
		97	1 second	Continuous
Cooling	1	37	30 seconds	None

Table 2.1: qPCR program

Every phase of the thermal cycling protocol has a particular purpose that is vital to the qPCR assay's accuracy and fidelity. At 95°C, the pre-incubation process activates the hot-start DNA polymerase and denatures the starting template strands. In each cycle's denaturation step, double-stranded DNA is separated so that primers can bind. During the annealing step, the primers can hybridize to their specific target sequences under controlled conditions. During the extension phase, Taq polymerase creates a complementary DNA strand that grows from the primer. The melting curve analysis performed at the end of the run measures the melting temperature of the amplicon to ensure amplification specificity; Single distinct peaks denote specific products, whereas multiple or low-temperature peaks may indicate nonspecific products or primer dimers. Lastly, the cooling step serves to stabilize the reaction components and readies the plate for removal and additional analysis.

2.4.3 Use of SYBR Green Dye

In this assay, SYBR Green I served as the intercalating dye. Upon excitation, it emits a strong fluorescent signal and binds specifically to the minor groove of double-stranded DNA (dsDNA). The fluorescence rises in proportion to the accumulation of PCR product during each cycle, enabling the real-time observation of amplification. Compared to probe-based chemistries, SYBR Green is a more economical and simpler detection system; however, it necessitates a melting curve analysis for specificity confirmation, as it attaches to any dsDNA—including primer dimers and nonspecific products (Arya et al., 2005). Our experiments demonstrated the specificity of amplification for both GAPDH and PARKIN through the occurrence of distinct, sharp melting peaks.

3.Results and Interpretation

3.1 DNA Quantification and Sample Selection (Nanodrop)

Using the Thermo Scientific™ Nanodrop™ spectrophotometer, the DNA content and purity of all 48 patient and control samples were evaluated. Using absorbance ratios at A260/A280 and A260/A230 wavelengths, a volume of 1 μ L from each sample was analyzed to assess the nucleic acid content (ng/ μ L) and quality. The majority of samples displayed A260/A280 ratios at 1.8–2.0, indicating good purity and minimal protein contamination, as can be shown in the whole dataset (Appendix/Table X).

A group of eight samples was chosen based on their similar DNA concentrations, which fell between 40 and 50 ng/ μ L, in order to guarantee uniformity and fairness in the subsequent qPCR analysis. The samples listed below were selected:

- **Patient group:** H2 (49.5), H9 (45.4), H15 (46.7), H20 (47.1)
- **Control group:** K6 (52.4), K14 (47.2), K15 (49.3), K24 (46.3)

By using this method, we were able to minimize variability unrelated to biological variations and preserve consistency in reaction efficiency and template input. Every sample that was chosen was suitably diluted to provide a constant amount of DNA input for each response.

	A	B	C	D	E	F	G	H
1	Sample	DNA Concentration ng/ μ L	Volume for 100ng (μ L)	Volume for 50ng (μ L)	A260	A280	260/280	260/230
2	H1	61.2	1.63	0.82	1.224	0.619	1.98	0.89
3	H2	49.5	2.02	1.01	0.991	0.512	1.94	0.97
4	H3	111	0.9	0.45	2.22	1.159	1.92	1.34
5	H4	30.2	3.31	1.66	0.605	0.335	1.8	0.86
6	H5	43.2	2.31	1.16	0.864	0.456	1.89	1.05
7	H6	45.5	2.2	1.1	0.911	0.476	1.91	1.06
8	H7	55.2	1.81	0.91	1.104	0.579	1.91	1.26
9	H8	38	2.63	1.32	0.761	0.399	1.91	1.03
10	H9	45.4	2.2	1.1	0.909	0.498	1.83	0.95
11	H10	41.3	2.42	1.21	0.826	0.453	1.82	1.04
12	H11	26.4	3.79	1.89	0.528	0.282	1.87	0.77
13	H12	43.3	2.31	1.15	0.866	0.475	1.82	0.9
14	H13	85.1	1.18	0.59	1.701	0.894	1.9	1.44
15	H14	103	0.97	0.49	2.06	1.093	1.88	1.69
16	H15	46.7	2.14	1.07	0.934	0.479	1.95	1.18
17	H16	55.3	1.81	0.9	1.106	0.575	1.92	1.2
18	H17	51.7	1.93	0.97	1.033	0.535	1.93	1.27
19	H18	54.4	1.84	0.92	1.088	0.564	1.93	1.35
20	H19	21.5	4.65	2.33	0.429	0.22	1.95	0.84
21	H20	47.4	2.11	1.05	0.974	0.488	1.94	1.15
22	H21	64.8	1.54	0.77	1.296	0.677	1.91	1.41
23	H22	38.3	2.61	1.31	0.765	0.392	1.95	1.12
24	H23	88.1	1.14	0.57	1.762	0.929	1.9	1.5
25	H24	102.1	0.98	0.49	2.042	1.08	1.89	1.56
26	H25	66.9	1.49	0.75	1.338	0.696	1.92	1.31

Figure 3.0: patients nanodrop results

	A	B	C	D	E	F	G	H
27 K1		73.9	1.35	0.68	1.479	0.776	1.91	1.31
28 K2		72.5	1.38	0.69	1.451	0.771	1.88	1.09
29 K3		40.7	2.46	1.23	0.813	0.416	1.95	1.04
30 K4		30.2	3.31	1.66	1.692	0.896	1.89	1.45
31 K5		60.9	1.64	0.82	1.218	0.638	1.91	1.32
32 K6		52.4	1.91	0.95	1.049	0.536	1.96	1.29
33 K7		42.6	2.35	1.17	0.852	0.444	1.92	1.19
34 K8		43.1	2.32	1.16	0.836	0.452	1.91	1.2
35 K9		42	2.38	1.19	0.84	0.445	1.89	1.16
36 K10		38.8	2.58	1.29	0.775	0.402	1.93	1.05
37 K11		78.9	1.27	0.63	1.578	0.821	1.92	1.45
38 K12		67.4	1.48	0.74	1.348	0.708	1.9	1.44
39 K14		47.2	2.12	1.06	0.945	0.498	1.9	1.23
40 K15		49.3	2.03	1.01	0.986	0.516	1.91	1.29
41 K16		81.2	1.23	0.62	1.625	0.853	1.9	1.6
42 K17		42	2.38	1.19	0.84	0.436	1.92	1.21
43 K18		33.2	3.01	1.51	0.663	0.34	1.95	1.02
44 K19		41.3	2.42	1.21	0.826	0.423	1.95	1.05
45 K20		33	3.03	1.52	0.66	0.341	1.94	0.98
46 K21		41.5	2.41	1.2	0.831	0.437	1.9	1.16
47 K22		36.7	2.72	1.36	0.735	0.38	1.94	1.12
48 K23		34.3	2.92	1.46	0.685	0.352	1.95	1.06
49 K24		46.3	2.16	1.08	0.927	0.484	1.92	1.12
50 K25		34.4	2.91	1.45	0.688	0.406	1.69	0.78

Figure 3.1: control nanodrop results

3.2 qPCR Results Analysis

The mean ΔCt was lower in the patient group (0.5025) compared to controls (0.955), indicating higher PARKIN expression in patients relative to GAPDH. The fold changes ($2^{-\Delta\Delta Ct}$) for patients ranged between 1.038 and 2.077, while controls ranged between 0.79 and 1.12.

Based on the reference scale (see Figure):

- $<0.10 \rightarrow$ zero copies
- $0.30-0.70 \rightarrow$ one copy
- $0.80-1.20 \rightarrow$ two copies
- $>1.20-1.80 \rightarrow$ three copies
- Values in $0.11-0.29$ or $0.71-0.79$ are considered intermediate.

The findings imply that PARKIN gene expression is somewhat elevated in certain patient samples, especially H2 and H20, which might be the result of an increase in gene dosage (copy number ~ 3). This can be a sign of underlying genetic variation or a compensatory reaction. About two copies of variance may be seen in the remaining patient samples, some of which are in intermediate zones and should be interpreted with caution.

On the other hand, control samples demonstrate constant PARKIN expression levels under typical circumstances, primarily falling within the anticipated two-copy range. It is possible for one control (K15) to exhibit mild underexpression, approaching an intermediate one-copy condition.

While these findings suggest potential upregulation of PARKIN in certain patients, further confirmation using a larger sample size, replicate runs, or probe-based qPCR would strengthen the conclusions.

	A	B	C	D	E	F	G	H	I	J
1	Patient	GAPDH (CT)	PRKN (CT)	ΔCt		ΔΔCt (H)		2 ^{ΔΔCt}	2 ^{ΔΔCt}	
2	H2	23.29	23.19	-0.1		-1.055		0.4813	2.07772	3 copies
3	H9	23.17	23.95	0.78		-0.175		0.88577	1.12896	2 copies
4	H15	22.73	23.63	0.9		-0.055		0.96259	1.03886	2 copies
5	H20	22.79	23.22	0.43		-0.525		0.69496	1.43893	3 copies
6			mean	0.5025						
7										
8	Control	GAPDH(CT)	PRKN(CT)	ΔCt		ΔΔCt (k)		2 ^{ΔΔCt}	2 ^{ΔΔCt}	
9	K6	22.24	23.13	0.89		-0.065		0.95595	1.04608	2 copies
10	K14	22.26	23.12	0.86		-0.095		0.93627	1.06807	2 copies
11	K15	21.93	23.22	1.29		0.335		1.26138	0.79278	1-2 copies
12	K24	22.45	23.23	0.78		-0.175		0.88577	1.12896	2 copies
13			mean	0.955						
14										
15									Reference Scale	
16									<0.1	0 copies
17									0.3-0.7	1 copy
18									0.8-1.2	2 copies
19									>1.2-1.8	3 copies

Figure 3.2: ΔCt and ΔΔCt analysis of PARKIN gene expression in patient and control samples

3.2.1 Ct Calculations

- ΔCt (Delta Ct) was calculated for each sample as:

$$\Delta Ct = Ct_{PARKIN} - Ct_{GAPDH}$$

This normalizes PARKIN expression to the housekeeping gene GAPDH.

- ΔΔCt (Delta-Delta Ct) was calculated by comparing each patient's ΔCt to the average ΔCt of the control group:

$$\Delta\Delta Ct = \Delta Ct_{\text{sample}} - \Delta Ct_{\text{control avg.}}$$

- The fold change in expression (relative expression) was determined using the formula:

$$\text{Fold Change} = 2^{-\Delta\Delta Ct}$$

This approach allows for the comparison of relative gene expression levels between patient and control groups, accounting for technical and biological variability.

4. Discussion

The purpose of this study was to investigate the possible relationship between PARKIN (PARK2) expression levels and male infertility. PARKIN is a gene that is known to be involved in paternal mitochondrial elimination (PME). In order to guarantee maternal mitochondrial inheritance, PME, a crucial post-fertilization pathway, preferentially degrades the paternal mitochondria. Because mitochondrial retention from sperm might have a detrimental effect on embryonic development or mitochondrial genome integrity, disruption of this mechanism may be a contributing factor to male infertility.

We compared the expression of PARKIN to that of GAPDH, a widely used housekeeping gene, in eight samples—four from infertile patients and four from fertile controls—to see if there were any differences in PARKIN expression in male infertiles. All of the individuals' DNA had already been extracted, and a selection was selected based on high purity ratios and comparable quantities (40–50 ng/μL), which were verified by Nanodrop spectrophotometry. This reduced technological variability and guaranteed constant qPCR input.

Both GAPDH and PARKIN primers were premade and supplied in dry, desalted form. To get final usable quantities, nuclease-free water was added. This was followed by dilution to 10 μM and then to 2.5 μM working solutions for qPCR. Using tools like NCBI, Ensembl, reverse complement checking, and UCSC In-Silico PCR, GAPDH primers underwent comprehensive bioinformatic validation, revealing that they amplified a spliced transcript region with an ideal product size of 139 bp. On the other hand, PARKIN primers were effectively employed in experiments, demonstrating strong amplification, but their inability to match entire transcript sequences prevented definitive validation.



Using the Roche LightCycler® 96 platform and SYBR Green chemistry, the qPCR experiment was carried out. Rows A–D represented patient samples, whereas rows E–H represented controls. The GAPDH and PARKIN responses were loaded into columns 3 and 8, respectively. Heat maps and amplification curves verified consistent and positive amplification in every sample.

Ct values were used to calculate ΔCt (PARKIN – GAPDH), $\Delta\Delta\text{Ct}$ (relative to control average), and fold changes using the $2^{-\Delta\Delta\text{Ct}}$ method. The mean ΔCt in patient samples was lower than in controls (0.5025 vs. 0.955), indicating a modest increase in PARKIN expression. Specifically, two patients (H2 and H20) exhibited a $2^{-\Delta\Delta\text{Ct}}$ value of 2.07772 and 1.43893 respectively, corresponding to three gene copies based on established interpretation thresholds, while the others clustered around two copies. In comparison, most controls had stable two-copy expression, with one (K15) falling into an intermediate zone (0.79), potentially indicating slight underexpression.

These findings imply that certain male infertiles have slightly elevated PARKIN expression, with at least one instance possibly involving copy number variation. This might be a cellular compensatory response to mitochondrial malfunction or compromised mitophagy. Because PARKIN and PINK1 are essential for marking paternal mitochondria for autophagic destruction, their dysregulation may interfere with PME and lead to improper zygotic development or unsuccessful fertilization.

The existence of a possible copy number increase and persistent upregulation in patients indicates the importance of PARKIN in male infertility linked to faulty mitochondrial clearance, even though the expression difference between patients and controls was not significant. However, these results should be regarded with caution because of the small sample size and the poor validation of PARKIN primers.



5. Conclusion

In summary, this study offers preliminary evidence that certain male infertiles have altered PARKIN expression, which may indicate a connection between reproductive failure and defective paternal mitochondrial elimination (PME). PARKIN's upregulation and copy number increase might be a sign of molecular compensation for abnormalities in mitochondrial clearance. These results emphasize that in order to validate PARKIN's involvement in male infertility and its potential as a diagnostic biomarker, bigger cohort studies, protein-level validation, and functional tests are required.

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